Senior Cryptography Engineer

Applied Cryptography, Secure Systems & Privacy-Preserving Technologies

📍 Remote / Hybrid / Onsite
🕒 Full-Time or Part-Time
🏢 The Copia Group, LLC
🔐 Senior Level

About The Copia Group

The Copia Group is an investment management firm supporting innovative companies across blockchain, cybersecurity, fintech, AI, and emerging technology sectors.

Through strategic investments and partnerships, we help accelerate the development of secure digital infrastructure, decentralized technologies, financial platforms, and privacy-focused software solutions. Our portfolio companies are building technologies where security, trust, and cryptographic innovation are fundamental to their success.

We are seeking an experienced **Senior Cryptography Engineer** to design, implement, and optimize cryptographic systems that protect sensitive data, secure digital assets, and enable trusted distributed applications.

The Opportunity

As a Senior Cryptography Engineer, you will lead the design and implementation of advanced cryptographic solutions across blockchain platforms, secure communication systems, authentication services, and privacy-preserving technologies. You'll collaborate with Security Engineers, Software Engineers, Blockchain Architects, and Product teams to build secure, scalable, and production-ready systems.

This role offers the opportunity to solve complex security challenges while contributing to next-generation technologies including digital identity, decentralized systems, confidential computing, and zero-knowledge cryptography.

What You'll Do

Applied Cryptography

- Design and implement modern cryptographic protocols and secure communication systems
- Develop encryption, digital signature, authentication, and key management solutions
- Build secure cryptographic libraries and reusable security components
- Evaluate emerging cryptographic standards and recommend implementation strategies
- Perform cryptographic threat modeling and security analysis

Blockchain & Distributed Systems

- Develop cryptographic solutions supporting blockchain infrastructure and decentralized applications
- Implement secure wallet, transaction signing, and digital asset management systems
- Design cryptographic protocols for consensus mechanisms and distributed networks
- Collaborate with blockchain teams on protocol security and smart contract integration
- Improve security across Web3 and decentralized ecosystems

Privacy & Security Engineering

- Build privacy-preserving technologies using advanced cryptographic techniques
- Implement secure authentication, authorization, and identity verification systems
- Design secure key lifecycle management and Hardware Security Module (HSM) integrations
- Conduct security reviews, penetration support, and cryptographic audits
- Ensure compliance with industry security standards and best practices

Technical Leadership

- Provide guidance on cryptographic architecture and secure software development
- Mentor engineers on cryptographic principles and security engineering practices
- Collaborate with Product, Infrastructure, and Compliance teams throughout development
- Participate in architecture reviews and security assessments
- Stay current with academic research and advancements in applied cryptography

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What We're Looking For

Required Qualifications

Experience

- 5+ years of professional software engineering or security engineering experience
- 3+ years of hands-on experience designing or implementing cryptographic systems
- Experience building production-grade security solutions
- Strong understanding of secure software development practices

Technical Expertise

Strong experience with:

- Applied Cryptography
- Public Key Infrastructure (PKI)
- TLS/SSL
- Digital Signatures
- Key Management Systems (KMS)
- Hardware Security Modules (HSM)
- Secure Authentication Protocols
- C++, Rust, Go, or Python

Strong understanding of:

- Symmetric and asymmetric cryptography
- Elliptic Curve Cryptography (ECC)
- RSA and AES encryption
- Hash functions and message authentication
- Zero-Knowledge Proofs (ZKP)
- Multi-Party Computation (MPC)
- Threshold cryptography
- Secure random number generation

- Secure key exchange protocols
- Cryptographic attack vectors and mitigation strategies

Preferred Qualifications

● Experience building blockchain protocols, digital asset platforms, or cryptocurrency infrastructure

● Familiarity with zk-SNARKs, zk-STARKs, Bulletproofs, or other zero-knowledge technologies

● Experience with secure enclave technologies such as Intel SGX or AWS Nitro Enclaves

● Experience implementing FIDO2, WebAuthn, OAuth 2.0, OpenID Connect, or identity management systems

● Knowledge of post-quantum cryptography and emerging cryptographic standards

● Contributions to open-source cryptography libraries, blockchain projects, or academic research

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Technology Environment

Programming Languages
Rust • C++ • Go • Python • Java

Cryptography & Security
OpenSSL • libsodium • BoringSSL • PKI • HSM • KMS

Blockchain & Web3
Ethereum • Solidity • Bitcoin • Digital Signatures • MPC • Zero-Knowledge Proofs

Cloud & Infrastructure
AWS • Azure • Google Cloud Platform • Docker • Kubernetes

Security & Compliance
OAuth 2.0 • OpenID Connect • WebAuthn • FIDO2 • SOC 2 • ISO 27001

Tools & Collaboration
GitHub • Jira • Slack • Notion • Security Testing Tools

Why Join Us

✨ Flexible remote, hybrid, and onsite work options

✨ Full-time and part-time opportunities

✨ Competitive compensation package

✨ Opportunity to develop next-generation cryptographic technologies and secure digital infrastructure

✨ Exposure to innovative blockchain, cybersecurity, fintech, and AI initiatives

✨ Collaborative, security-first engineering culture focused on research and innovation

✨ Career advancement into Principal Cryptography Engineer, Security Architect, or Cryptography Research Lead

✨ Strong investment in technical research, conference participation, open-source contributions, and continuous learning

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Apply

If you are passionate about applied cryptography, secure system design, and building technologies that protect digital assets and sensitive information at scale, we'd love to connect with you.